

TG-40 (Monthly QA)

Wednesday, July 31, 2024 7:46 AM



TG40 TG142
TG198...

★ Monthly QA:

- Verify parameters that have a
 - Lower likelihood of changing on a day to day basis
 - Smaller impact on patient treatment quality
- Tolerances tighter than for daily QA, because
 - Dosimetry devices calibrated by Accredited Dosimetry Calibration Laboratory (ADCL)
 - Performed by QMP
- Best practice order of operations (gantry head → phantom)
 - Safety
 - Mechanical
 - Imaging
 - Dosimetry

• Monthly Mechanical

TABLE II. Monthly.

Procedure	Machine-type tolerance			
	Non-IMRT	IMRT	SRS/SBRT	
Mechanical				
Light/radiation field coincidence ^b		2 mm or 1% on a side		
Light/radiation field coincidence ^b (asymmetric)		1 mm or 1% on a side		<u>2 mm</u>
Distance check device for lasers compared with front pointer		1mm		<u>2 mm</u>
Gantry/collimator angle indicators (@ cardinal angles) (digital only)		1.0°		
Accessory trays (i.e., port film graticule tray)		2 mm		
Jaw position indicators (symmetric) ^c		2 mm		
Jaw position indicators (asymmetric) ^d		1 mm		<u>2 mm</u>
Cross-hair centering (walkout)		1 mm		<u>2 mm</u>
Treatment couch position indicators ^e	2 mm/1°	2 mm/1°	1 mm/0.5°	<u>2 mm/1°</u>
Wedge placement accuracy		2 mm		
Compensator placement accuracy ^f		1 mm		
Latching of wedges, blocking tray ^g		Functional		
Localizing lasers	±2 mm	±1 mm	< ±1 mm	<u>2 mm</u>

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• Monthly Dosimetry

TABLE II. Monthly.

Procedure	Machine-type tolerance			
	Non-IMRT	IMRT	SRS/SBRT	
Dosimetry				
X-ray output constancy				Same
Electron output constancy		2%		
Backup monitor chamber constancy				
Typical dose rate ^a output constancy	NA	2% (@ IMRT dose rate)	2% (@ stereo dose rate, MU)	
Photon beam profile constancy		1%		<u>2%</u>
Electron beam profile constancy		1%		<u>3%</u>
Electron beam energy constancy		2%/2 mm		<u>2mm</u>
				<u>2 %</u>

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• Monthly Safety/Gating

TABLE II. Monthly.

Machine-type tolerance				
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Procedure	Non-IMRT	IMRT	SRS/SBRT
Safety			
Laser guard-interlock test		Functional	
Respiratory gating			
Beam output constancy		2%	
Phase, amplitude beam control		Functional	
In-room respiratory monitoring system		Functional	
Gating interlock		Functional	
			None
			N/A

- **Monthly Imaging**

TABLE VI. Imaging.

Procedure	Application-type tolerance	
	non-SRS/SBRT	SRS/SBRT
Monthly		
Planar MV imaging (EPID)		
Imaging and treatment coordinate coincidence (four cardinal angles)	≤2 mm	≤1 mm
Scaling ^b	≤2 mm	≤2 mm
Spatial resolution	Baseline ^c	Baseline
Contrast	Baseline	Baseline
Uniformity and noise	Baseline	Baseline
Planar kV imaging^d		
Imaging and treatment coordinate coincidence (four cardinal angles)	≤2 mm	≤1 mm
Scaling	≤2 mm	≤1 mm
Spatial resolution	Baseline	Baseline
Contrast	Baseline	Baseline
Uniformity and noise	Baseline	Baseline
Cone-beam CT (kV and MV)		
Geometric distortion	≤2 mm	≤1 mm
Spatial resolution	Baseline	Baseline
Contrast	Baseline	Baseline
HU constancy	Baseline	Baseline
Uniformity and noise	Baseline	Baseline



2 mm

2.5 mm

Same

Same

Same

2 mm

Same

Same

Same

2 mm

Same

Same

Same

Same